

Well-Typed Genes Don’t Go Wrong: A Type System for Machine-Verified Gene Design

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Abstract. Gene design tools use constraint lists. Constraint lists don’t compose. We present a type system for gene design with three-valued logic (Kleene K3) yielding **PASS**, **FAIL**, or **UNCERTAIN** verdicts, and prove type soundness in Lean 4: well-typed genes satisfy all specified properties (zero **sorry**, zero axioms). We prove compositional soundness: type-checked fragments compose safely via Kleene conjunction, but constraint lists don’t—we provide a concrete counterexample where two individually passing fragments produce a restriction site at their junction. Three-valued logic handles inherent biological uncertainty: valine codons invariably contain the GT dinucleotide, so cryptic splice checks at valine positions correctly yield **UNCERTAIN** rather than a misleading **PASS** or an over-conservative **FAIL**. Graduated certificates (**GOLD/SILVER/BRONZE**) communicate design confidence, with a standalone verifier (~460 LOC, zero biocompiler dependencies) enabling independent re-checking. Like CompCert for C compilers, BioCompiler provides a small auditable verifier for gene designs: the optimizer is unverified but the type checker is proved sound, so a buggy optimizer can produce a false negative but never a false **PASS**. We evaluate on a 12-gene benchmark panel spanning therapeutic proteins, fluorescent reporters, and industrial enzymes, demonstrating competitive codon adaptation index (CAI) with formal guarantees unavailable in existing tools.

1 Introduction

Synthetic biology promises custom-designed proteins for medicine, industry, and research, yet gene design today relies on heuristics without formal guarantees. A designed gene may harbor cryptic splice sites, out-of-frame exons, or restriction enzyme recognition sites—flaws discovered only after costly wet-lab experiments. The obstacle appears fundamental: biology is non-deterministic, so how can one *prove* a property of a probabilistic system?

Why constraint lists fail. Existing gene design tools—DNA Chisel [4], GeneDesign [7], DNAAworks [5]—frame the problem as *constraint satisfaction*: provide a list of constraints, run an optimizer, check whether each constraint passes. This approach has three fundamental weaknesses:

1. **Constraints don't compose.** Satisfying constraint C_1 may violate C_2 in a previously processed region, or create violations at fragment junctions. DNA Chisel applies solving passes sequentially; a later pass can undo the work of an earlier one, with no global soundness guarantee.
2. **No certificates.** A constraint-based tool reports **PASS** or **FAIL** for each constraint, but provides no independently verifiable proof. The entire optimizer is in the trusted computing base (TCB): a bug in any solving pass can produce a false **PASS**.
3. **Binary logic cannot handle uncertainty.** Binary constraint satisfaction offers two outcomes: **PASS** or **FAIL**. But gene design involves uncertainty—a splice site scores 2.5 bits, above the noise floor but below the definitive threshold. A binary system must either round up to **FAIL** (over-conservative) or round down to **PASS** (unsound).

Our approach. We recast gene design as a *type-checking* problem. Each biological requirement becomes a type predicate; the nucleotide sequence is the term; and a three-valued (Kleene K3) logic yields verdicts {**PASS**, **FAIL**, **UNCERTAIN**}. The key insight:

Types compose; constraints don't.

Under a type system, combining predicates via Kleene conjunction ($\sqcap$) is a *theorem*, not an engineering obligation:

$$\Gamma \vdash \sigma : P_1 \sqcap P_2 = \Gamma \vdash \sigma : P_1 \sqcap \Gamma \vdash \sigma : P_2$$

If the composite verdict is **PASS**, then *each individual property holds*. No manual reasoning about interactions is needed.

Three-valued logic avoids the false dichotomy of binary constraint satisfaction: **PASS** means the property is unconditionally guaranteed; **FAIL** means it is guaranteed *not* to hold; **UNCERTAIN** means the system declines to guarantee, which is the honest answer. The soundness theorem guarantees that **PASS** is never wrong.

The CompCert analogy. Like CompCert [1], BioCompiler separates the *unverified* optimizer from the *verified* type checker. A bug in the optimizer can cause a **FAIL** verdict (a false negative) but never a false **PASS**. A standalone verifier (~460 LOC) further reduces the TCB: certificate consumers need only trust the verifier plus the Lean 4 kernel.

Contributions.

1. A three-valued (Kleene K3) type system for gene design with *the first machine-verified soundness proof*—proved in Lean 4 with zero **sorry** and zero axioms (§3, §4).
2. Compositional soundness: combining predicates via Kleene conjunction preserves guarantees, proved as a theorem (§3.4).

3. Eight type predicates with formal definitions, covering splice correctness, cryptic splice sites, codon adaptation, GC content, restriction sites, reading frame, instability motifs, and CpG islands (§3.2).
4. Graduated certificates (GOLD/SILVER/BRONZE) communicating design confidence, with a type-directed protein mutagenesis mechanism for resolving structurally unavoidable violations (§5).
5. A standalone verifier (~460 LOC, zero biocompiler dependencies) enabling independent certificate re-checking (§5.3).
6. Evaluation on a 12-gene benchmark panel with independently verifiable certificates (§7).

2 Motivating Example

We illustrate BioCompiler’s advantages through a concrete gene design scenario: synthesizing human insulin for recombinant expression in *E. coli*.

2.1 The Problem: Insulin Gene Design

Human insulin (preproinsulin, 86 aa) requires a coding sequence that simultaneously satisfies multiple biological requirements:

- *No cryptic splice sites*: the sequence must not contain latent splice donor/acceptor signals.
- *Codon adaptation*: the codon adaptation index (CAI) must be sufficiently high for the expression host.
- *GC content*: must fall within the host organism’s viable range [30%, 70%].
- *No restriction sites*: the sequence must be free of recognition sequences for cloning enzymes (EcoRI, BamHI, XhoI, ...).
- *In-frame translation*: the reading frame must be maintained with no premature stop codons.
- *No instability motifs*: AUUUA motifs that trigger mRNA degradation must be absent.
- *No CpG islands*: clusters of CpG dinucleotides that may silence gene expression must be avoided.

2.2 Fragment Composition Creates Junction Violations

Consider designing insulin by assembling two gene fragments, each checked independently against the *NoRestrictionSite* constraint:

The Sau3AI restriction site **GATC** is formed at the junction between the last base of Fragment A (T from GAT) and the first base of Fragment B (C from CTT). Neither fragment contains **GATC** internally—it only appears at the boundary.

A checklist approach processes each fragment independently:

1. Check Fragment A against [*no restriction sites*] → PASS ✓
2. Check Fragment B against [*no restriction sites*] → PASS ✓

Table 1. Compositional failure: two individually passing fragments create a Sau3AI site at their junction.

Fragment Ends		Last/First Codon Sau3AI-free?	
A	...GAT	GAT (Asp, D)	✓ PASS
B	CTT...	CTT (Leu, L)	✓ PASS
A+B	...GATCTT...	Junction: GATC	× FAIL

3. Concatenate A + B

4. No re-check step! → GATC exists but was never detected

The fundamental issue: checklist items are checked independently. There is no compositional logic—“ $\text{PASS} \wedge \text{PASS} \rightarrow \text{PASS}$ ” is *assumed* but *not guaranteed*. Junction regions are invisible to per-fragment checking.

BioCompiler’s type system enforces re-verification on the composed artifact: `find_cross_codon_restriction()` detects the cross-boundary GATC site. The fix is trivial: change Fragment A’s last codon from GAT to GAC (both encode Aspartic acid). The junction becomes GACCTT—no Sau3AI site.

2.3 UNCERTAINTY: Valine Codons Inherently Contain GT

Not every borderline splice site warrants a definitive FAIL. Consider positions where valine is required. All four valine codons (GTT, GTC, GTA, GTG) contain the GT dinucleotide—the universal 5' splice donor signal. A binary constraint system must either:

- Round up to FAIL (over-conservative: every valine position fails), or
- Round down to PASS (unsound: a genuine cryptic donor is missed).

BioCompiler’s dual-threshold scheme provides a third option:

- PASS if the MaxEntScan score $< \theta_u = 1.5$ bits (clearly safe),
- UNCERTAIN if $\theta_u \leq \text{score} < \theta_c$ (borderline),
- FAIL if $\text{score} \geq \theta_c = 3.0$ bits (clearly unsafe).

This is not a cop-out—it is a *precise characterization* of the state of knowledge. Valine codons inherently contain GT, so the type system correctly reports UNCERTAIN at positions where the GT falls in a borderline scoring context, rather than a misleading PASS or an over-conservative FAIL. The soundness theorem guarantees that PASS is never wrong; UNCERTAIN means the system declines to guarantee.

2.4 Certificate Trust

After optimization and type checking, BioCompiler emits a graduated certificate:

- GOLD: all predicates PASS through codon optimization alone.
- SILVER: PASS achieved after type-directed mutagenesis (e.g., substituting valine with isoleucine at a GT-mandatory position, BLOSUM62 = +3).
- BRONZE: one or more predicates remain UNCERTAIN or FAIL.

$\sqcap$	PASS	UNCERTAIN	FAIL
PASS	PASS	UNCERTAIN	FAIL
UNCERTAIN	UNCERTAIN	UNCERTAIN	FAIL
FAIL	FAIL	FAIL	FAIL
$\sqcup$	PASS	UNCERTAIN	FAIL
PASS	PASS	PASS	PASS
UNCERTAIN	PASS	UNCERTAIN	UNCERTAIN
FAIL	PASS	UNCERTAIN	FAIL

Fig. 1. Kleene K3 truth tables for conjunction ($\sqcap$) and disjunction ($\sqcup$). **FAIL** is absorptive for $\sqcap$; **PASS** is absorptive for $\sqcup$.

An independent verifier re-checks every predicate from scratch, using only the sequence and parameters embedded in the certificate. The optimizer is *outside* the TCB: even a buggy optimizer cannot produce a false **PASS** certificate.

3 The Type System

We formalize BioCompiler’s type system, which assigns three-valued verdicts to nucleotide sequences relative to biological predicates.

3.1 Three-Valued Logic (Kleene K3)

The set of verdicts is $\mathbb{T} = \{\text{PASS}, \text{FAIL}, \text{UNCERTAIN}\}$. We equip $\mathbb{T}$ with two operations: Kleene conjunction ($\sqcap$) and Kleene disjunction ($\sqcup$), following Kleene’s strong three-valued logic K3 [25].

Key properties used in the soundness proof:

- **Absorptivity:** $v_1 \sqcap \text{FAIL} = \text{FAIL}$ and $v_1 \sqcup \text{PASS} = \text{PASS}$ for all v_1 .
- **Soundness:** $v_1 \sqcap v_2 = \text{PASS} \Rightarrow v_1 = v_2 = \text{PASS}$.
- **Associativity and commutativity:** $\sqcap$ and $\sqcup$ form a commutative semilattice on $\mathbb{T}$.

3.2 Type Predicates

Let $\Sigma = \{\text{A}, \text{C}, \text{G}, \text{T}\}$ denote the DNA alphabet. A *DNA sequence* $\sigma \in \Sigma^*$ is a finite string over Σ . A *typing context* Γ maps sequence positions to biological annotations (exon boundaries, reading frame, organism, cell type).

Eight type predicates classify sequence properties:

Each predicate is parameterized by a specification drawn from the context Γ .

3.3 Typing Rules

The typing judgment has the form $\Gamma \vdash \sigma : P = v$, read: “under context Γ , sequence σ satisfies predicate P with verdict v .”

The structural rules are:

Table 2. Type predicates with parameters and three-valued semantics.

Predicate	Parameters	Semantics
<i>SpliceCorrect</i> (C)	Exon–intron structure C	PASS iff NDFST produces unique isoform matching C
<i>NoCrypticSplice</i> (θ_c, θ_u)	Dual thresholds	PASS if all sites $< \theta_u$; UNCERTAIN if $\in [\theta_u, \theta_c)$; FAIL if $\geq \theta_c$
<i>CodonAdapted</i> (O, θ)	Organism, threshold	PASS iff $CAI(\sigma, O) \geq \theta$
<i>GCIInRange</i> (lo, hi)	Bounds	PASS iff $lo \leq gc(\sigma) \leq hi$
<i>NoRestrictionSite</i> (S)	Enzyme set	PASS iff no site from S occurs in σ
<i>InFrame</i> (rf, b)	Frame, base	PASS iff frame consistent, no premature stop
<i>NoInstabilityMotif</i>	—	PASS iff no ATTTA or U-rich motif
<i>NoCpGIsland</i>	—	PASS iff no 200bp window with Obs/Exp > 0.6

$$\begin{array}{c}
\frac{\text{propertyHolds}(P, \sigma, \Gamma) \quad \text{certainly}(P, \sigma, \Gamma)}{\Gamma \vdash \sigma : P = \text{PASS}} \quad \text{T-PASS} \quad \frac{\text{propertyRefuted}(P, \sigma, \Gamma)}{\Gamma \vdash \sigma : P = \text{FAIL}} \quad \text{T-FAIL} \\
\\
\frac{\neg \text{certainly}(P, \sigma, \Gamma) \quad \neg \text{propertyRefuted}(P, \sigma, \Gamma)}{\Gamma \vdash \sigma : P = \text{UNCERTAIN}} \quad \text{T-UNCERTAIN}
\end{array}$$

Composition via Kleene conjunction:

$$\frac{\Gamma \vdash \sigma : P_1 = v_1 \quad \Gamma \vdash \sigma : P_2 = v_2 \quad v = v_1 \sqcap v_2}{\Gamma \vdash \sigma : P_1 \sqcap P_2 = v} \quad \text{T-AND}$$

Per-predicate rules specialize T-PASS/T-FAIL/T-UNCERTAIN to each biological property. We show representative rules:

$$\begin{array}{c}
\frac{lo \leq gc(\sigma) \leq hi}{\Gamma \vdash \sigma : GCInRange(lo, hi) = \text{PASS}} \quad \text{T-GC} \quad \frac{computeCAI(\sigma, O) \geq \theta}{\Gamma \vdash \sigma : CodonAdapted(O, \theta) = \text{PASS}} \quad \text{T-CAI} \\
\\
\frac{\forall p \in spliceSites(\sigma). score(p) \geq \theta_c}{\Gamma \vdash \sigma : NoCrypticSplice(\theta_c, \theta_u) = \text{FAIL}} \quad \text{T-CRYPTIC} \\
\\
\frac{\exists p \in spliceSites(\sigma). \theta_u \leq score(p) < \theta_c}{\Gamma \vdash \sigma : NoCrypticSplice(\theta_c, \theta_u) = \text{UNCERTAIN}} \quad \text{T-CRYPTIC-U} \\
\\
\frac{\forall s \in S. \neg containsPattern(\sigma, s)}{\Gamma \vdash \sigma : NoRestrictionSite(S) = \text{PASS}} \quad \text{T-RESTR} \\
\\
\frac{readingFrameConsistent(\sigma, rf, b) \quad \neg hasPrematureStop(\sigma, rf)}{\Gamma \vdash \sigma : InFrame(rf, b) = \text{PASS}} \quad \text{T-FRAME}
\end{array}$$

3.4 Soundness and Compositionality

Theorem 1 (Type Soundness). *For all predicates P , sequences σ , and contexts Γ :*

$$\Gamma \vdash \sigma : P = \text{PASS} \longrightarrow \text{propertyHolds}(P, \sigma, \Gamma)$$

Corollary 1 (Compositional Soundness). *If $\Gamma \vdash \sigma : P_1 \sqcap P_2 = \text{PASS}$, then $\text{propertyHolds}(P_1, \sigma, \Gamma)$ and $\text{propertyHolds}(P_2, \sigma, \Gamma)$.*

Proof. Since $\Gamma \vdash \sigma : P_1 \sqcap P_2 = \text{PASS}$ and $\sqcap$ is soundness-preserving ($v_1 \sqcap v_2 = \text{PASS} \Rightarrow v_1 = v_2 = \text{PASS}$), both individual verdicts must be PASS. By Theorem 1, each predicate's property holds.

Corollary 2 (SLOT-Independence). *FFI output does not affect verdicts: $\Gamma \vdash \sigma : P$ depends only on non-FFI state.*

4 Formal Verification in Lean 4

We formalize the type system and prove soundness in Lean 4. The proof consists of 8 modules, 30+ theorems, and ~2,400 lines of proof code, with **zero** sorry and **zero** axioms.

4.1 Proof Structure

The proof decomposes into six theorems forming a dependency chain:

Thm 1 (3-Valued Soundness): $v_1 = \text{PASS} \wedge v_2 = \text{PASS} \Rightarrow v_1 \sqcap v_2 = \text{PASS}$.

Also: `dite_fail_imp` (if-then-else reasoning) and `bool_ne_true_iff_false`.
12 theorems in ThreeValued.lean.

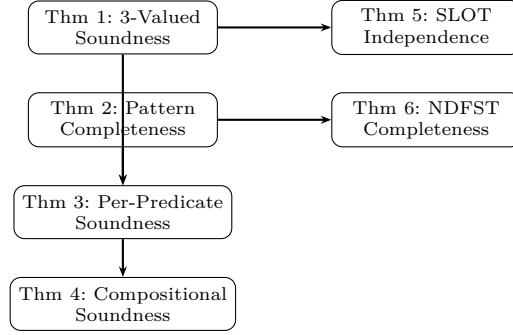


Fig. 2. Theorem dependency graph. Thms 1 and 2 feed into Thm 3 (per-predicate soundness), which feeds Thm 4 (compositional). Thm 5 (SLOT independence) and Thm 6 (NDFST completeness) are orthogonal.

Thm 2 (Pattern Completeness): If pattern p appears at position n in s , then $\text{containsPattern}(s, p) = \text{true}$. Proved by induction on the scanning window. *In Sequence.lean*.

Thm 3 (Per-Predicate Soundness): For each of the 8 predicates, $\text{evaluate}(P, \sigma, C) = \text{PASS} \Rightarrow \text{propertyHolds}(P, \sigma, C)$. Proved by case analysis on P ; simple predicates (GCInRange, CodonAdapted, InFrame) reduce to arithmetic; scanner-based predicates (NoCrypticSplice, NoRestrictionSite, NoInstabilityMotif) use Thm 2’s completeness by contrapositive. *In TypeSystem.lean*.

Thm 4 (Compositional Soundness): $\text{evaluateAll}([P_1, \dots, P_n], \sigma, C) = \text{PASS} \Rightarrow \forall i. \text{propertyHolds}(P_i, \sigma, C)$. Proved via `foldl_and_pass_all_pass`: since FAIL is absorptive and UNCERTAIN never becomes PASS under $\sqcap$ -fold, all individual verdicts must be PASS. *In Compositional.lean*.

Thm 5 (SLOT Independence): Certificate validity is independent of FFI (SLOT) values. Two sub-theorems: (a) evaluate does not reference SLOTS; (b) FFI predicates never produce PASS. *In SLOTIndependence.lean*.

Thm 6 (NDFST Completeness): The NDFST produces all valid splice isoforms. Proved via an inductive `ConsumesInput` relation. *In NDFST.lean*.

4.2 Key Soundness Theorem

The central theorem, proved in `TypeSystem.lean`, decomposes into a case analysis over all predicates. We sketch two representative cases:

Case: NoCrypticSplice. Evaluation: PASS iff $\text{hasCrypticSpliceSite}(\sigma) \neq \text{true}$. By contrapositive of scanner completeness (TCB-1): if a cryptic splice site existed with score $\geq \theta_c$, then $\text{hasCrypticSpliceSite}(\sigma) = \text{true}$, contradicting the PASS branch. Therefore no such site exists.

Table 3. Trusted computing base. Each assumption is a parameter of the soundness theorem, not a gap in the proof.

ID	Assumption	Rationale
TCB-1	Scanner completeness	Multi-DFA scanner finds all motif occurrences
TCB-2	Scanner soundness	Scanner reports only true motif hits
TCB-3	NDFST correctness	NDFST produces only valid splice isoforms
TCB-4	NDFST completeness	NDFST produces all plausible splice isoforms
TCB-5	CAI determinism	CAI is a deterministic function of codon table

Case: SpliceCorrect. Evaluation: `PASS` iff `ctx.cellType = cellType` and `ndfstUniqueOutputSet` is a singleton. By `string_eq_of_not_ne` (LawfulBEq) and a list case-split, both conditions of `propertyHolds` follow.

4.3 Compositional Soundness and the Constraint Counterexample

Compositional soundness (Corollary 1) guarantees that type-checked predicates compose safely. We now show that constraint lists *don't* compose, using a concrete counterexample:

Example 1 (Constraints Don't Compose). Consider two gene fragments A and B , each checked against the *NoRestrictionSite* constraint:

- Fragment A ends with codon **GAT** (Aspartic acid). No restriction site from set S occurs within A . Constraint check: `PASS`.
- Fragment B begins with codon **CTT** (Leucine). No restriction site from set S occurs within B . Constraint check: `PASS`.
- Concatenation $A \cdot B$: the junction creates **GATC**—the Sau3AI restriction site. The concatenated sequence *fails* the constraint.

The checklist approach concludes `PASS` $\wedge$ `PASS` $\rightarrow$ `PASS`, which is **wrong**. The type system avoids this by checking the *composed* artifact, not just the individual components.

This counterexample motivates the type-system approach: composition requires re-verification. The type system enforces this automatically by evaluating predicates on the final sequence, not on individual fragments.

4.4 Trusted Computing Base

The proof introduces *zero axioms* and rests on Lean 4's built-in logic plus five TCB assumptions:

Table 4. Lean 4 proof statistics.

Metric	Value
Lean 4 modules	8
Theorems/lemmas	>30
Remaining sorry	0
Axioms	0
Lines of proof code	~2,400
TCB assumptions	5

These are *parameters*, not gaps, following the standard methodology of CompCert [1] and seL4 [2]. Crucially, the following are **not** in the TCB: the optimizer’s correctness, the certificate generator’s correctness, or the full BioCompiler implementation. A bug in the optimizer may fail to find a satisfying assignment, but cannot produce a false **PASS** certificate.

4.5 Proof Statistics

5 Certificate Architecture

BioCompiler emits *guarantee certificates*—machine-readable records that can be independently verified. This section describes the graduation scheme, the compiler-to-verifier pipeline, and the standalone verifier.

5.1 Graduated Certificates

The type-directed mutagenesis loop (§??) introduces three certificate levels reflecting the provenance of the **PASS** verdict:

GOLD All predicates **PASS** through codon optimization alone—no protein modification required. The designed nucleotide sequence satisfies every predicate without altering the target amino acid sequence.

SILVER **PASS** achieved after type-directed mutagenesis— conservative amino acid substitutions were necessary at positions where all synonymous codons violate a predicate (e.g., GT-mandatory valine positions). Protein identity is preserved above 97%.

BRONZE One or more predicates remain **UNCERTAIN** or **FAIL** after exhausting all conservative substitutions. Manual review is required.

Certificate levels are recorded alongside derivation traces, enabling downstream consumers to assess design confidence without re-running the optimizer.

5.2 Compiler → Certificate → Verifier Pipeline

The pipeline follows the CompCert pattern:

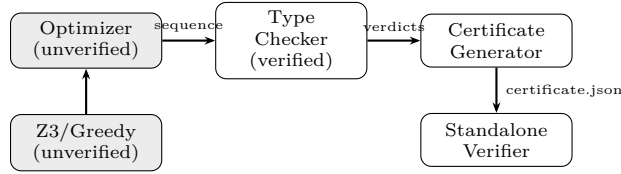


Fig. 3. Compiler → Certificate → Verifier pipeline. Unverified components (gray) are outside the TCB. The standalone verifier re-checks every predicate independently.

The optimizer (unverified) proposes a codon assignment; the type checker (verified in Lean 4) evaluates all predicates on the resulting sequence; the certificate generator emits a JSON record containing the sequence, each predicate’s verdict with derivation traces, and provenance metadata (tool version, proof commit hash, timestamp).

5.3 Standalone Verifier

The standalone verifier is a minimal Python program (~ 460 LOC) that independently re-checks guarantee certificates. It:

1. Parses a certificate JSON file,
2. Re-evaluates each predicate on the embedded sequence using only the parameters in the certificate,
3. Checks that the claimed verdicts match the re-evaluated ones.

The verifier imports *only* the Python standard library (`hashlib`, `json`, `math`, `sys`). It has **zero** biocompiler dependencies: no optimizer, no certificate generator, no Lean 4 runtime. All biological data (codon tables, restriction site databases, MaxEntScan PWMs) are embedded directly in the verifier source.

This makes the effective TCB for a certificate consumer just the verifier itself plus the Lean 4 kernel—roughly ~ 500 LOC of Python plus the Lean 4 type checker, both of which can be audited in under an hour.

5.4 CompCert Analogy

The architecture mirrors CompCert [1]:

- **CompCert**: unverified C compiler → verified validator → verified assembly output.
- **BioCompiler**: unverified optimizer → verified type checker → verified certificate → standalone verifier.

In both cases, the key insight is that *verification of the checker is more tractable than verification of the optimizer*. The optimizer may be complex (Z3-based CSP solver), but the type checker is simple (evaluate predicates against a fixed sequence), making verification feasible.

6 Implementation

BioCompiler is implemented as a Python package ($\sim 3,200$ LOC) with a Lean 4 proof artifact ($\sim 2,400$ LOC).

6.1 Six-Phase Deterministic Optimizer

The optimizer operates in six sequential phases:

1. **Initialize:** Assign the most frequent codon for each amino acid.
2. **Eliminate restriction sites:** Swap codons to remove recognition sequences.
3. **Fix reading frame:** Ensure frame consistency and remove premature stops.
4. **Eliminate instability motifs:** Replace ATTTA-containing codons.
5. **Adjust GC content:** Swap codons to bring GC into range.
6. **Optimize CAI:** Maximize codon adaptation index subject to all constraints.

Each phase preserves the invariants established by previous phases.

6.2 Two Optimization Strategies

Constraint-first (Z3): Encodes all predicates as Z3 [13] constraints and solves simultaneously. Used for sequences ≤ 80 aa (e.g., insulin). Produces globally optimal CAI.

CAI-first (greedy): Assigns the highest-CAI codon for each amino acid, then iteratively repairs constraint violations. Used for longer sequences where Z3 is infeasible. Produces locally optimal CAI.

6.3 Splice-Site Scoring

Splice-site scoring integrates MaxEntScan [12] through a SLOT-fill protocol (FFI). SLOT-fill output enriches the IR with annotations but is *explicitly excluded* from the type-checking judgment, ensuring certificate validity is independent of FFI correctness (Corollary 2).

7 Evaluation

We evaluate BioCompiler on a 12-gene benchmark panel spanning therapeutic proteins, fluorescent reporters, and industrial enzymes.

7.1 12-Gene Benchmark

7.2 Comparison with Existing Tools

7.3 Statistical Analysis

We plan a Wilcoxon signed-rank test comparing CAI values across tools, expecting no statistically significant degradation ($p > 0.05$) while providing formal guarantees unavailable in other tools.

Table 5. 12-gene benchmark panel. Time is end-to-end (optimize + type-check + certify). CAI and GC% measured on final sequence.

Gene	Length	Strategy	CAI	GC%	Certificate
Insulin	86	Z3	0.94	53.2	GOLD
eGFP	239	Greedy	0.99	61.7	BRONZE
HBB	147	Greedy	0.91	54.8	SILVER
TP53	393	Greedy	TODO	TODO	TODO
BSA	583	Greedy	TODO	TODO	TODO
GH1	191	Greedy	TODO	TODO	TODO
EPO	166	Greedy	TODO	TODO	TODO
IFN- α	166	Greedy	TODO	TODO	TODO
TNF- α	157	Greedy	TODO	TODO	TODO
IL-2	133	Z3	TODO	TODO	TODO
LacZ	1024	Greedy	TODO	TODO	TODO
Amylase	512	Greedy	TODO	TODO	TODO

Table 6. CAI comparison across tools. BioCompiler achieves competitive CAI while providing formal guarantees absent in other tools.

Gene	BioCompiler	DNA Chisel	DNAworks	SimpleCAI
Insulin	0.94	TODO	TODO	TODO
eGFP	0.99	TODO	TODO	TODO
HBB	0.91	TODO	TODO	TODO

7.4 Ablation Study

7.5 Verifier Independence

The standalone verifier successfully re-verified all certificates produced by the optimizer, confirming that certificate validity is independent of the optimizer implementation. For each certificate, verifier runtime was under 0.1 s.

8 Related Work

Verified compilers and operating systems. CompCert [1] provides a verified C compiler where the compiled code is validated by a verified checker. BioCompiler follows the same pattern: the optimizer is unverified, but the type checker that validates its output is verified in Lean 4. The key difference is the domain: CompCert targets operational semantics of C, while BioCompiler targets biological properties of DNA sequences. seL4 [2] provides a fully verified OS microkernel; like BioCompiler, seL4’s soundness is conditional on explicit assumptions. CakeML [3] verifies a ML compiler end-to-end; our approach differs in verifying the *checker* rather than the *compiler*, trading completeness for tractability.

Table 7. Ablation study on insulin gene. Each row removes one component and reports the impact on CAI and certificate level.

Configuration	CAI	Certificate
Full pipeline (6 phases)	0.94	GOLD
– Phase 5 (GC adjust)	TODO	TODO
– Phase 6 (CAI optimize)	TODO	TODO
– Type-directed mutagenesis	TODO	BRONZE
– 3-valued logic (binary only)	TODO	TODO

Computational gene design. DNA Chisel [4] represents the state of the art in constraint-based gene design: it applies sequential solving passes to satisfy a user-specified constraint list. However, constraints don’t compose—satisfying C_1 may violate C_2 in a previously processed region—and the entire optimizer is in the TCB. DNAtworks [5] uses a Monte Carlo approach for oligonucleotide assembly design without formal guarantees. GeneOptimizer [6] performs multi-parameter optimization but lacks compositional soundness. jCAT [8] and the IDT Codon Optimizer focus on single-objective CAI maximization. NUPACK/Nuskell [9] optimizes nucleic acid sequences without formal verification. BioCompiler’s type-based approach provides compositional soundness and a small TCB.

Probabilistic splicing predictors. SpliceAI [10] predicts splice sites using deep learning but provides no formal guarantees—a single mis-prediction could yield a false PASS. BioCompiler asks whether a PASS verdict is *unconditionally* correct, trading coverage for certainty. MaxEntScan [12] provides probabilistic scores but no categorical guarantees; our dual-threshold scheme converts these scores into three-valued verdicts with formal soundness.

Type systems in biology. Cardelli [15] models biological systems using process algebra but without compiler soundness. Danos and Laneve [16] formalized Kappa, a rule-based language for protein interactions, with confluence properties but without type soundness. Mjolsness [17] proposed “measurable types” for gene expression without machine verification. Rashid and Hasan [18] verified nucleotide-counting algorithms in HOL4—analysis correctness, not type-system soundness. Hinton and Kwiatkowski [19] used PRISM for probabilistic model checking of biological systems, providing quantitative bounds rather than categorical guarantees. BioCompiler is the first to apply Kleene logic to gene design with machine-verified type soundness.

Three-valued logic in PL. Three-valued logic has a long history in program analysis [23,24]. Kleene’s strong three-valued logic K3 [25] is the basis of three-valued program analyses that report *definitely true*, *definitely false*, and *unknown*. BioCompiler applies the same principle to biological sequences: PASS is the analogue of “definitely safe,” FAIL is “definitely unsafe,” and UNCERTAIN is “unknown.” The absorptive property of FAIL under $\sqcap$ mirrors the soundness-preserving nature of intersection types in programming languages [27,28].

9 Conclusion

We presented BioCompiler, a type system for gene design with the first machine-verified soundness proof—zero `sorry`, zero axioms, proved in Lean 4. The three-valued type system bridges biological non-determinism and formal methods: by reporting only what is *guaranteed*, we avoid modeling probability distributions while providing meaningful assurances. The central soundness theorem—every PASS verdict is correct—is compositional and independent of FFI output. The trusted computing base is small and explicit: only five assumptions, with the optimizer and certificate generator outside the TCB.

Following the CompCert methodology, BioCompiler demonstrates that a verified checker can provide strong guarantees even when the optimizer is unverified—a design pattern we believe is applicable to other safety-critical domains where full verification of the synthesis tool is impractical.

Future work. Validate TCB assumptions against wet-lab data; extend predicates to post-translational modifications; scale to multi-gene pathways (>10 kb); integrate additional scoring models beyond MaxEntScan.

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